

COMPUTER VISION (IT524)

Objective:

Broad topics include discussions on approaches to 3D (object distance from camera) estimation using 2D image pixels. Face recognition. Basics of deep learning and its use in high resolution image generation, image generation from random noise, etc.

Pre-requisites :

Good Understanding of:

- Fundamentals of Digital Image Processing
- Linear Algebra
- Probability and Random variables
- Programming skills

Topics:

Brief ideas on image processing, Geometric transformation (Affine, Projective), Camera Models, Projective Geometry, Image formation, Camera matrix, Camera Calibration, Shape from stereo (stereopsis), correspondence problem, Ideas on estimation techniques LS (least squares), CLS (constrained least squares), Maximum likelihood (ML) estimation, Maximum A posteriori (MAP) estimation, super-resolution imaging, Principal Component Analysis (PCA) for face recognition, Motion Field and Optical flow, Concepts of MRF (Markov random fields), introduction to neural networks, Generative models (GANs).

Outcome: The course would help the graduate students to understand the basic concepts in computer vision (how the computer can be used to perform the tasks of human vision system). Through lectures, programming assignments and presentations the students are provided with the learning experience of

1. Analysis of images for extracting the 3D coordinates of the object

2. Develop technical presentation skills
3. Solving the problems that are of ill-posed nature.

Text Books:

1. Multiple view geometry in computer vision: R. Hartley and A. Zisserman, Cambridge University Press
2. Robot Vision by BKP Horn, MIT Press
3. Introductory Techniques for 3-D Computer Vision by Trucco and Veri, Prentice Hall
4. Deep Learning: Ian Goodfellow, Yashua Bengio and Aaron Courville, MIT Press.

Evaluation Strategy:

In semester and Mid semester Exams: 40% each

Best of the above 2 exams will be considered.

Project: 50%

Attendance, Class participation, Assignments: 10%

MAP TO PROGRAM OUTCOMES (POs) OF NBA

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X					X	X		X